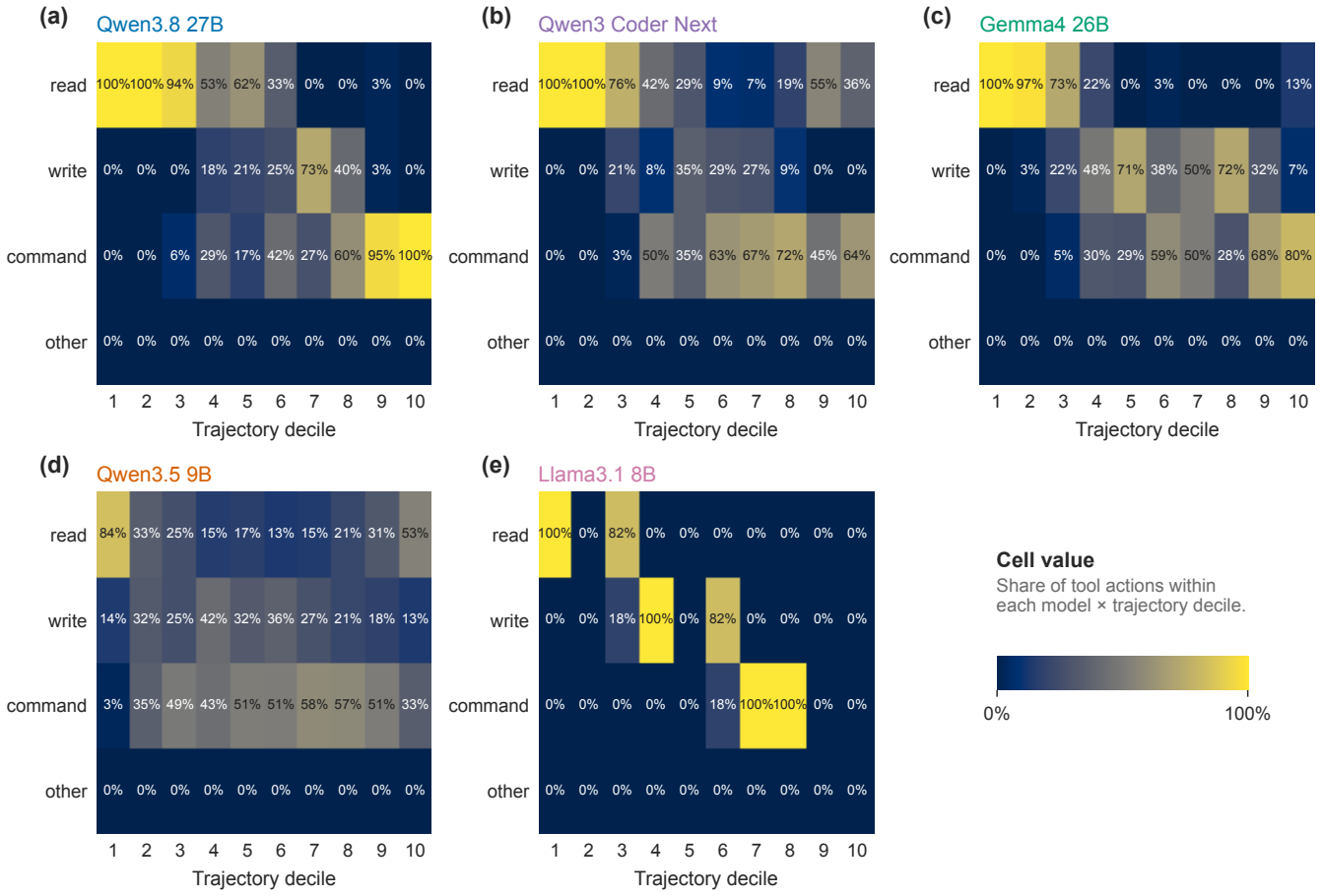
(a)
Qwen3.8 27B
read 100% 100% 94% 53% 62% 33% 0% 0% 3% 0%
write 0% 0% 0% 18% 21% 25% 73% 40% 3% 0%
command 0% 0% 6% 29% 17% 42% 27% 60% 95% 100%
other 0% 0% 0% 0% 0% 0% 0% 0% 0% 0%
1 2 3 4 5 6 7 8 9 10
Trajectory decile
(b)
Qwen3 Coder Next
read 100% 100% 76% 42% 29% 9% 7% 19% 55% 36%
write 0% 0% 21% 8% 35% 29% 27% 9% 0% 0%
command 0% 0% 3% 50% 35% 63% 67% 72% 45% 64%
other 0% 0% 0% 0% 0% 0% 0% 0% 0% 0%
1 2 3 4 5 6 7 8 9 10
Trajectory decile
(c)
Gemma4 26B
read 100% 97% 73% 22% 0% 3% 0% 0% 0% 13%
write 0% 3% 22% 48% 71% 38% 50% 72% 32% 7%
command 0% 0% 5% 30% 29% 59% 50% 28% 68% 80%
other 0% 0% 0% 0% 0% 0% 0% 0% 0% 0%
1 2 3 4 5 6 7 8 9 10
Trajectory decile
(d)
Qwen3.5 9B
read 84% 33% 25% 15% 17% 13% 15% 21% 31% 53%
write 14% 32% 25% 42% 32% 36% 27% 21% 18% 13%
command 3% 35% 49% 43% 51% 51% 58% 57% 51% 33%
other 0% 0% 0% 0% 0% 0% 0% 0% 0% 0%
1 2 3 4 5 6 7 8 9 10
Trajectory decile
(e)
Llama3.1 8B
read 100% 0% 82% 0% 0% 0% 0% 0% 0% 0%
write 0% 0% 18% 100% 0% 82% 0% 0% 0% 0%
command 0% 0% 0% 0% 0% 18% 100% 100% 0% 0%
other 0% 0% 0% 0% 0% 0% 0% 0% 0% 0%
1 2 3 4 5 6 7 8 9 10
Trajectory decile
Cell value
Share of tool actions within each model × trajectory decile.
0%
100%